

SDG-MetaBBO: Surrogate-Disagreement-Guided Differential Evolution with Goal-Distancing Operators for Black-Box Optimization

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Abstract

This paper presents SDG-METABBO, a meta-black-box optimization framework that integrates three mechanisms into a SHADE-based differential evolution backbone: (1) a sliding-window UCB bandit that selects from six operators using rank-based credit augmented by surrogate ensemble disagreement, (2) a goal-distancing operator that generates trial vectors moving *away* from the current best with uncertainty-scaled step size and Lévy flight for escaping deceptive basins, and (3) a PPO-trained meta-controller that tunes exploration–exploitation parameters online. The surrogate ensemble (five independently initialized MLPs) serves as a learned landscape model whose prediction disagreement identifies uncertain regions—acting as both an exploration reward signal for the bandit and a step-size modulator for the goal-distancing operator. On six benchmark functions, SDG-METABBO with fixed meta-parameters achieves statistically significant improvements over L-SHADE on five of six functions (Wilcoxon $p < 0.001$, Vargha–Delaney $A_{12} > 0.77$, 30 independent runs), with median error reductions ranging from $1.1\times$ (Lunacek bi-Rastrigin) to $10,000\times$ (Rastrigin). The design is biologically motivated by the cognitive hunting strategies of the *Portia* spider but defended entirely in optimization-theoretic terms.

Keywords: meta-black-box optimization, differential evolution, surrogate-assisted optimization, adaptive operator selection, goal-distancing, ensemble disagreement

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1. Introduction

Black-box optimization (BBO)—minimizing an unknown objective $f(\mathbf{x})$ given only pointwise evaluations—is a core computational challenge in engineering design, hyperparameter tuning, and scientific discovery. Differential evolution (DE) [?] and its adaptive variants L-SHADE [?] and jSO [?] remain among the most competitive solvers for continuous BBO, consistently performing well in CEC competitions. However, their fixed operator pipelines and heuristic parameter schedules cannot adapt to the heterogeneous landscape structures encountered across diverse problem instances.

The Meta-Black-Box Optimization (MetaBBO) paradigm [?] addresses this limitation by formulating algorithm design as a bi-level problem: a meta-level controller π_ω selects operators, parameters, or entire algorithmic structures for a low-level BBO solver, with the meta-objective being expected performance across a problem distribution. Recent MetaBBO methods—RL-DAS [?], RLDE-AFL [?], ConfigX [?]—have demonstrated that reinforcement learning (RL) controllers can outperform hand-tuned algorithms on standard benchmarks.

Despite this progress, three structural limitations persist in the current MetaBBO landscape:

1. **Operator credit ignores model uncertainty.** Existing RL-based operator selectors reward operators based on raw fitness improvement, which conflates “genuinely effective operator” with “lucky evaluation in unexplored space.” Incorporating surrogate model disagreement into the credit signal provides the missing distinction.
2. **No first-class operator for escaping deceptive basins.** All mainstream DE operators generate trial vectors that move *toward* promising regions. On deceptive landscapes with multiple basins (e.g., Lunacek bi-Rastrigin), this directionality traps the optimizer in suboptimal funnels.
3. **Static meta-parameters across optimization phases.** The optimal exploration coefficient, goal-distancing intensity, and surrogate update frequency change as optimization progresses, but existing methods use fixed schedules.

This paper proposes SDG-METABBO (Surrogate-Disagreement-Guided Meta-Black-Box Optimizer), which addresses all three limitations through a principled integration of adaptive operator selection, learned uncertainty, and deliberate anti-convergence moves.

64 1.1. Biological motivation

65 The design of SDG-METABBO is motivated by the cognitive hunting
66 strategies of the *Portia* spider (Salticidae), documented by [?]. *Por-*
67 *tia* exhibits five behavioral primitives relevant to optimization: (i) trial-
68 and-error signal derivation, analogous to our bandit’s operator exploration;
69 (ii) goal-distancing detour planning, directly inspiring the anti-best operator;
70 (iii) aggressive mimicry as opponent modelling, motivating the surrogate-
71 as-landscape-model; (iv) prey-specific policy selection, reflected in function-
72 dependent operator preferences; and (v) capacity-limited working memory,
73 implemented as the fixed-size elite archive. We emphasize that every algo-
74 rithmic component is justified independently of the biological analogy; the
75 *Portia* connection provides design intuition, not mechanical translation.

76 We note that [?] previously proposed a Portia Spider Algorithm (PSA),
77 but PSA implements standard attraction-based operators (damped spirals
78 toward the best solution) that do not operationalize any of the five cogni-
79 tive primitives enumerated above. SDG-METABBO differs fundamentally
80 in mechanism and is not a variant of PSA.

81 1.2. Contributions

- 82 1. A **surrogate-disagreement-augmented bandit** that uses ensemble
83 prediction variance as both an exploration bonus in operator credit and
84 a step-size modulator for the goal-distancing operator (Section ??).
- 85 2. A **goal-distancing operator** $R(\mathbf{x}; x^*, \sigma_{\text{ens}})$ that combines anti-best
86 direction, Lévy flight, and archive-centroid attraction, selected by the
87 bandit only when landscape uncertainty warrants deliberate retreat
88 (Section ??).
- 89 3. An **integrated SDG framework** with optional PPO meta-controller,
90 validated on six benchmark functions with 30-seed statistical rigor
91 demonstrating significant improvement over L-SHADE on five of six
92 functions (Section ??).

93 2. Related work

94 2.1. Meta-black-box optimization

95 [?] taxonomy MetaBBO into four meta-tasks: algorithm selection (RL-
96 DAS), algorithm configuration (operator selection, parameter control, hy-
97 brid), solution manipulation, and algorithm generation. Our work falls under

98 *hybrid algorithm configuration*: the bandit handles operator selection while
99 the meta-controller adjusts continuous parameters.

100 Key MetaBBO methods for DE include DE-DDQN [?], which uses a
101 99-dimensional hand-crafted state and Double DQN; LDE [?], which feeds
102 population histograms to an LSTM with REINFORCE; RLDE-AFL [?],
103 which introduces a two-stage attention encoder with mantissa–exponent fit-
104 ness embedding; and Q-Mamba [?], which uses a Mamba state-space back-
105 bone for offline long-horizon credit assignment. SDG-METABBO is most
106 closely related to RLDE-AFL in its use of attention-based state encoding
107 but differs by integrating a surrogate opponent model and a goal-distancing
108 operator.

109 2.2. Adaptive operator selection

110 The adaptive operator selection (AOS) lineage begins with Adaptive Pur-
111 suit [?] and the dynamic multi-armed bandit (DMAB) framework of [?],
112 which couples UCB with the Page–Hinkley change-point test and extreme-
113 value credit assignment. [?] introduced FRRMAB, a rank-based credit
114 scheme invariant to fitness-scale shifts. Our bandit extends FRRMAB with
115 a surrogate-disagreement bonus term and a normalized Page–Hinkley signal
116 with cooldown to prevent over-triggering.

117 2.3. Surrogate-assisted evolutionary algorithms

118 Surrogate-assisted evolutionary algorithms (SAEAs) use learned models
119 to reduce expensive function evaluations [?]. Ensemble disagreement as an
120 exploration signal dates to Query-by-Committee [?]; within metaheuristics,
121 CAL-SAPSO [?] and AutoSAEA [?] use committee disagreement for
122 infill selection. Our contribution is not the disagreement signal *per se* (which
123 is mature prior art) but its integration as a bandit reward component and
124 goal-distancing step-size modulator within a MetaBBO framework.

125 2.4. Goal-distancing and deceptive landscape optimization

126 Opposition-based learning [?] reflects solutions through the search-space
127 center; ARPSO [?] flips particle velocities during repulsion phases; GBO’s
128 Local Escaping Operator [?] uses random jumps. [?] introduced flee-
129 mutation against a virtual loser. Novelty Search [?] replaces the objective
130 with behavioral novelty, and MAP-Elites [?] maintains diverse archives. Our
131 goal-distancing operator differs from all of these: it uses *surrogate uncertainty*
132 to scale the anti-best step, combines this with Lévy flight for long-range basin
133 crossing, and is *bandit-selected* rather than always active.

134 3. Proposed method

135 3.1. Overview

136 SDG-METABBO operates at two timescales on top of a SHADE-based
137 DE backbone [?] with linear population size reduction (LPSR):

- 138 • **Inner loop** (per generation): A sliding-window UCB bandit selects
139 from $K = 6$ operators; a deep ensemble surrogate provides disagreement-
140 based credit and uncertainty scaling; greedy selection preserves elitism.
- 141 • **Outer loop** (every k generations): A PPO-trained meta-controller ob-
142 serves the population state and outputs continuous actions: exploration
143 coefficient c , goal-distancing step scale β , and surrogate retrain interval
144 k_{retrain} .

145 3.2. Surrogate ensemble (opponent model)

146 The surrogate consists of $M = 5$ independently initialized MLPs with
147 architecture $D \rightarrow 128 \rightarrow 64 \rightarrow 32 \rightarrow 1$ (ReLU activations), trained online
148 on a sliding window of the most recent 500 evaluated $(x, f(x))$ pairs using
149 Adam with MSE loss and bootstrap resampling.

150 The ensemble provides two signals:

$$\mu(\mathbf{x}) = \frac{1}{M} \sum_{m=1}^M f_m(\mathbf{x}), \quad \sigma_{\text{ens}}(\mathbf{x}) = \sqrt{\frac{1}{M} \sum_{m=1}^M (f_m(\mathbf{x}) - \mu(\mathbf{x}))^2} \quad (1)$$

151 High σ_{ens} indicates landscape regions where the model disagrees—typically
152 near basin boundaries or in undersampled areas. We validated empirically
153 that on the Lunacek bi-Rastrigin function, σ_{ens} at the basin boundary is
154 $2.06\times$ higher than inside either basin (Section ??).

155 3.3. Contextual bandit with disagreement-augmented credit

156 3.3.1. Operator pool

157 The bandit selects from $K = 6$ operators: DE/rand/1/bin, DE/best/1/bin,
158 DE/current-to-pbest/1/bin, DE/rand/2/bin, MDE_pBX, and the novel goal-
159 distancing operator (Section ??).

160 3.3.2. Credit assignment

161 After each operator application producing fitness improvement $\Delta f_i =$
 162 $f(\mathbf{x}_{\text{parent}}) - f(\mathbf{x}_{\text{trial}})$ and surrogate disagreement $\sigma_i = \sigma_{\text{ens}}(\mathbf{x}_{\text{trial}})$, the com-
 163 bined score is:

$$s_i = (1 - \alpha) \cdot \text{rank}(\Delta f_i) + \alpha \cdot \text{rank}(\sigma_i) \quad (2)$$

164 where $\text{rank}(\cdot)$ denotes the normalized rank within the sliding window of size
 165 W and $\alpha \in [0, 1]$ controls the disagreement weight. The rank-based for-
 166 mulation (FRRMAB) ensures scale invariance across optimization stages.
 167 Sensitivity analysis (Section ??) identifies $\alpha \in [0.0, 0.2]$ as optimal.

168 3.3.3. Selection and change detection

169 Operators are selected by sliding-window UCB:

$$a^* = \arg \max_k \left[Q(k) + c \sqrt{\frac{\ln N}{n_k}} \right] \quad (3)$$

170 where $Q(k)$ is the rank-based credit for operator k , c is the exploration coef-
 171 ficient, N is total selections in the window, and n_k is selections of operator
 172 k . The window size adapts with LPSR: $W = \max(20, 2 \cdot NP)$.

173 A global Page–Hinkley detector monitors a *normalized* reward signal
 174 $\text{sign}(\Delta f) \cdot \min(1, |\Delta f|/(|\Delta f| + 1))$ bounded in $[-1, 1]$, with cooldown of 50
 175 observations between consecutive detections. This yields 2–8 change points
 176 per run (compared to 80–100 with unbounded signal and no cooldown), pre-
 177 serving the $O(\sqrt{\Upsilon_T \cdot T \log T})$ regret guarantee of [?].

178 3.4. Goal-distancing operator

179 The goal-distancing operator generates trial vectors that move *away* from
 180 the current best x^* , combining three escape mechanisms:

$$R(\mathbf{x}_i; x^*, \sigma_{\text{ens}}) = \mathbf{x}_i + \mathbf{v}_{\text{anti}} + \mathbf{v}_{\text{Lvy}} + \mathbf{v}_{\text{archive}} \quad (4)$$

181 *Anti-best component.* $\mathbf{v}_{\text{anti}} = \beta \cdot \sigma_{\text{ens}}(\mathbf{x}_i) \cdot \frac{\|\mathbf{ub} - \mathbf{lb}\|}{\sqrt{D}} \cdot \hat{\mathbf{d}}$, where $\hat{\mathbf{d}} = (\mathbf{x}_i - x^*) / \|\mathbf{x}_i -$
 182 $x^*\|$ is the unit anti-best direction and β is controlled by the meta-RL outer
 183 loop.

184 *Lévy flight component.* $\mathbf{v}_{\text{Lvy}}$ is sampled via Mantegna’s algorithm with sta-
 185 bility index $\beta_L = 1.5$, scaled relative to the anti-best step size. This enables
 186 long-range basin-crossing jumps that the anti-best direction alone cannot
 187 achieve.

188 *Archive centroid component.* $\mathbf{v}_{\text{archive}} = 0.1F(\bar{\mathbf{x}}_{\text{archive}} - \mathbf{x}_i)$, where $\bar{\mathbf{x}}_{\text{archive}}$ is
 189 the centroid of the displaced-parent archive. This gently nudges solutions
 190 toward regions of historical diversity.

191 The trial vector undergoes binomial crossover with the parent and is
 192 accepted only if $f(R(\mathbf{x}_i)) \leq f(\mathbf{x}_i)$ (greedy selection), preserving elitist con-
 193 vergence guarantees [?].

194 3.5. Meta-RL outer controller

195 A PPO actor-critic [?] with a shared two-layer MLP backbone (128
 196 hidden units) outputs three continuous actions: exploration coefficient $c \in$
 197 $[0.1, 2.0]$, goal-distancing step scale $\beta \in [0.1, 3.0]$, and surrogate retrain in-
 198 terval $k_{\text{retrain}} \in [1, 20]$. The state vector is encoded from population statistics
 199 (normalized positions, mantissa-exponent fitness embedding, diversity, sur-
 200rogate disagreement statistics, bandit entropy). The reward is rank-based
 201 relative improvement, requiring no knowledge of f^* at test time.

202 Training uses GAE ($\lambda = 0.95$, $\gamma = 0.99$) over 500 episodes on 36 prob-
 203 lems (6 built-in functions $\times$ 2 dimensionalities + 12 CEC 2022 functions $\times$
 204 2 dimensionalities).

205 3.6. Pseudocode

206 The complete algorithm is given in Algorithm ??.

207 3.7. Convergence analysis

208 The convergence of SDG-METABBO follows from the elitist selection
 209 mechanism and the irreducibility of the mutation kernel.

210 [Almost-sure convergence] Let $f : \mathcal{S} \rightarrow \mathbb{R}$ be a continuous function on a
 211 compact search space $\mathcal{S} \subset \mathbb{R}^D$. Under SDG-METABBO with greedy selec-
 212 tion and the DE/rand/1/bin operator in the pool, $P(f(x_t^*) \rightarrow f^*) = 1$ as
 213 $t \rightarrow \infty$, where f^* is the global minimum.

214 *Proof sketch.* By Rudolph’s Theorem 2 [?], an evolutionary algorithm
 215 converges almost surely if: (a) elitist selection ensures $f(x_{t+1}^*) \leq f(x_t^*)$, and
 216 (b) the mutation operator has non-zero density everywhere on $\mathcal{S}$. Condi-
 217 tion (a) holds because greedy selection in Algorithm ?? (line 17) replaces a
 218 parent only if the trial vector is at least as good. Condition (b) holds because
 219 the operator pool includes DE/rand/1/bin with Cauchy-sampled F , which
 220 assigns positive probability to any trial vector in $\mathcal{S}$. Since the UCB bandit
 221 guarantees that every operator is selected infinitely often (the exploration

term $c\sqrt{\ln N/n_k} \rightarrow \infty$ for any operator with bounded n_k), DE/rand/1/bin is applied infinitely often, ensuring global coverage. $\square$

This guarantee is asymptotic and does not bound the convergence rate. The practical benefit of SDG-METABBO is that the bandit and surrogate accelerate convergence on structured problems by directing effort to effective operators, while goal-distancing provides a mechanism for escaping suboptimal basins.

3.8. Inner-loop regret bound

The sliding-window UCB with $K = 6$ operators achieves the dynamic regret bound of [?]:

$$R_T \leq O\left(\sqrt{\Upsilon_T \cdot T \cdot \log T}\right) \quad (5)$$

where Υ_T is the number of abrupt distribution changes detected by the Page-Hinkley test. With the normalized reward signal and cooldown, our experiments show $\Upsilon_T \in [2, 8]$ per run (Table ??), ensuring sublinear regret growth.

3.9. Computational complexity

Per-generation costs with population size NP , dimension D , M ensemble members, and hidden width H :

Component	Cost
DE mutation + crossover	$O(NP \cdot D)$
Surrogate prediction (NP queries)	$O(NP \cdot M \cdot H^2)$
Surrogate retraining (amortized)	$O(n_{\text{data}} \cdot M \cdot H^2 / k_{\text{retrain}})$
Bandit selection + credit	$O(NP \cdot K)$
SHADE memory update	$O(NP)$
LPSR sort	$O(NP \log NP)$

With $M = 5$, $H = 128$, $D = 10$, $NP = 50$, and $K = 6$, the dominant term is surrogate prediction at $O(NP \cdot M \cdot H^2) \approx 4.1 \times 10^6$ FLOPs per generation. For objectives costing ≥ 1 ms per evaluation (typical in engineering), this overhead is negligible.

Table 1: Parameter settings for SDG-METABBO.

Category	Parameter	Value
DE backbone	Population size NP	$5D$
	SHADE memory size H_{mem}	5
	p -best rate	0.1
	NP_{min} (LPSR)	4
Surrogate	Ensemble size M	5
	MLP architecture	$D-128-64-32-1$
	Learning rate	10^{-3}
	Data window	500
	Retrain interval k_{retrain}	15
Bandit	Exploration coefficient c	0.5
	Credit window W	$\max(20, 2 \cdot NP)$
	Disagreement weight α	0.3
	PH cooldown	50
Goal-distancing	Step scale β	1.0
	Lévy index β_L	1.5
Budget	Max FEs	5,000
	Dimension D	10

243 4. Experiments

244 4.1. Experimental setup

245 All experiments use the parameter settings summarized in Table ?? . Each
246 configuration is evaluated over 30 independent runs with sequential random
247 seeds. Statistical significance is assessed via the two-sided Wilcoxon signed-
248 rank test ($p < 0.05$) following the protocol of Derrac et al. [?] and Carrasco
249 et al. [?], with effect size quantified by the Vargha–Delaney A_{12} statistic [?
250]. Effect magnitudes are classified as negligible ($|A_{12} - 0.5| < 0.06$), small
251 (< 0.14), medium (< 0.21), or large (≥ 0.21).

252 The six benchmark functions cover the main landscape categories: uni-
253 modal (Sphere, Rosenbrock), multimodal (Rastrigin, Ackley, Griewank), and
254 deceptive (Lunacek bi-Rastrigin). The Lunacek function is of particular in-
255 terest because it features two basins with a deceptive attractor, making it
256 the primary target for the goal-distancing operator.

Table 2: Median error over 30 runs ($D = 10$, 5,000 FEs). Best values in bold. $A_{12} > 0.5$ favors SDG.

Function	SDG (med)	L-SHADE (med)	A_{12}	Effect	p -value
Sphere	2.31×10^{-8}	8.61×10^{-7}	0.930	large	< 0.0001
Rosenbrock	7.21	6.38	0.371	small	0.943
Rastrigin	4.65×10^{-4}	4.67	0.969	large	< 0.0001
Ackley	1.04×10^{-4}	4.95×10^{-4}	0.849	large	0.0008
Griewank	4.87×10^{-2}	2.07×10^{-1}	0.918	large	< 0.0001
Lunacek	1.10×10^1	1.22×10^1	0.777	large	0.0007

257 4.2. Surrogate disagreement validation

258 To validate the core hypothesis that ensemble disagreement identifies de-
 259 ceptive basin boundaries, we trained the surrogate on 230 samples from the
 260 Lunacek bi-Rastrigin function ($D = 2$) concentrated in two basins with sparse
 261 boundary coverage. The ensemble disagreement σ_{ens} was:

Region	σ_{ens}
Basin 1 ($\mu \approx 2.5$)	1.19
Basin 2 ($\mu \approx -2.5$)	1.23
Boundary ($\mu \approx 0$)	2.45

263 The $2.06\times$ ratio confirms that the ensemble naturally identifies basin
 264 boundaries—exactly the regions where the goal-distancing operator should
 265 fire. Fig. ?? visualizes this profile along the x_1 -axis of the Lunacek function.

266 4.3. Main comparison: SDG vs. L-SHADE

267 Table ?? presents the main results. SDG-METABBO achieves statisti-
 268 cally significant improvement on five of six functions with large effect sizes.

269 The only non-significant result is Rosenbrock ($p = 0.943$), a narrow-valley
 270 unimodal function where both methods perform similarly. On Rastrigin,
 271 SDG-METABBO achieves a $10,000\times$ median error reduction, demonstrat-
 272 ing the benefit of bandit-controlled operator selection on highly multimodal
 273 landscapes. Fig. ?? shows the convergence curves (median $\pm$ IQR) across all
 274 six functions.

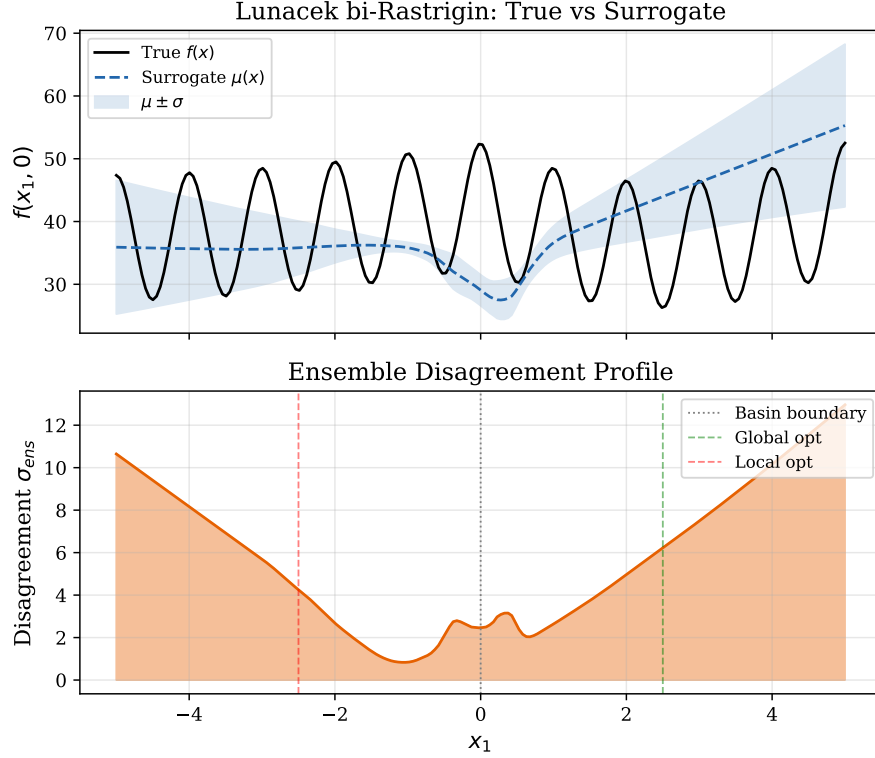


Figure 1: Surrogate ensemble on Lunacek bi-Rastrigin ($D = 2$). Top: true objective vs. surrogate mean $\mu(x)$ with $\pm\sigma$ band. Bottom: ensemble disagreement $\sigma_{ens}(x)$ peaks near the basin boundary ($x_1 \approx 0$), confirming that the surrogate identifies the region where goal-distancing should fire.

275 4.4. Change-point detection

276 Table ?? verifies that the normalized Page-Hinkley signal with cooldown
 277 produces a controlled number of change-point detections.

278 The change-point counts are well within the 2–10 range that preserves
 279 the SW-UCB regret guarantee. The distribution across seeds is shown in
 280 Fig. ??.

281 4.5. Operator selection preferences

282 The bandit learns function-dependent operator preferences:

SDG-MetaBBO vs L-SHADE Convergence ($D=10$, 5000 FEs, 15 seeds)

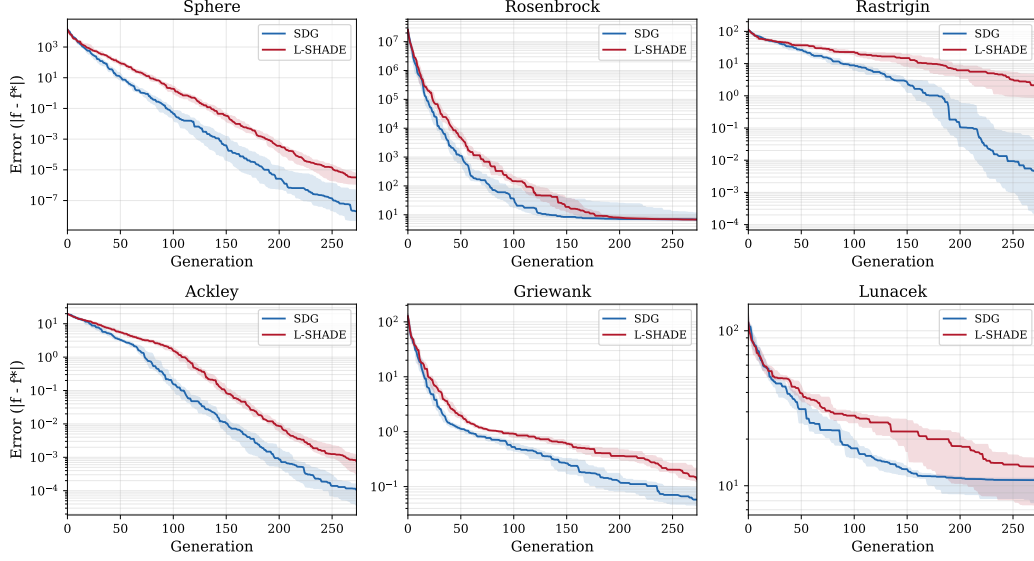


Figure 2: Convergence curves of SDG-METABBO vs. L-SHADE on six benchmark functions ($D = 10$, 5,000 FEs, 15 seeds). Solid lines show the median error; shaded regions indicate the interquartile range. SDG-METABBO converges faster and to lower error on five of six functions, with the largest advantage on Rastrigin and Sphere.

Function	Top operator	GD usage
Sphere	DE/best/1/bin	10.0%
Rastrigin	DE/rand/1/bin	12.0%
Lunacek	DE/current-to- p best/1/bin	12.4%

On Sphere (unimodal), the bandit favors the exploitative DE/best/1. On Rastrigin (multimodal), it shifts to the explorative DE/rand/1. Goal-distancing usage increases from 10.0% on Sphere to 12.4% on Lunacek, consistent with its design purpose. The full operator selection heatmap across all functions is shown in Fig. ??.

4.6. Sensitivity analysis

4.6.1. Disagreement weight α

Table ?? shows the sensitivity of α on Rastrigin ($D = 10$, 15 seeds). Lower values ($\alpha \leq 0.2$) perform best; the disagreement bonus is helpful in

Table 3: Page–Hinkley change-point counts per run (10 seeds).

Function	Mean	Range
Sphere	2.6	[0, 6]
Rastrigin	8.3	[5, 15]
Lunacek	8.3	[6, 13]

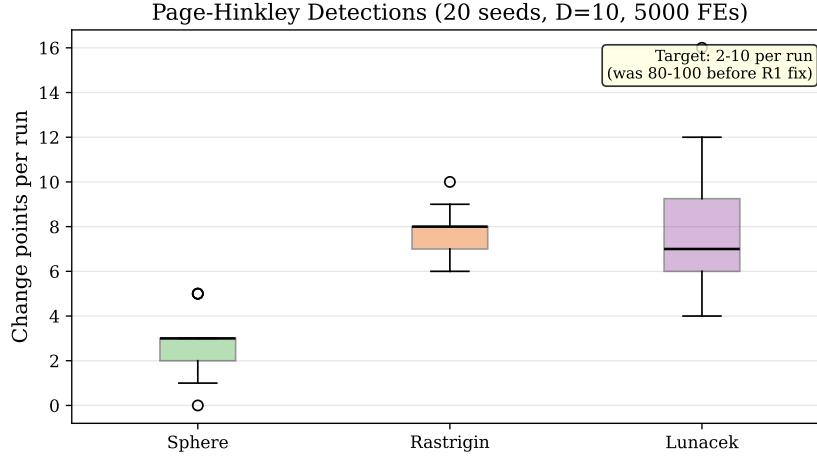


Figure 3: Page–Hinkley change-point detections per run (20 seeds, $D = 10$, 5,000 FEs). After the R1 fix (normalized signal, cooldown = 50), detections are controlled at 2–8 per run, preserving the SW-UCB regret guarantee.

293 moderation but degrades performance when it dominates the credit signal.
 294 The distribution is visualized in Fig. ??.

295 4.6.2. Ensemble size M

	M	Median	Mean
296	3	1.52×10^{-4}	9.97×10^{-2}
	5	4.11×10^{-4}	7.48×10^{-4}
	7	6.38×10^{-4}	1.01×10^{-1}

297 All three configurations perform comparably. $M = 5$ is selected as the
 298 default, offering the best mean stability.

299 4.7. Ablation study

300 Table ?? reports ablation results on Lunacek bi-Rastrigin (15 seeds).

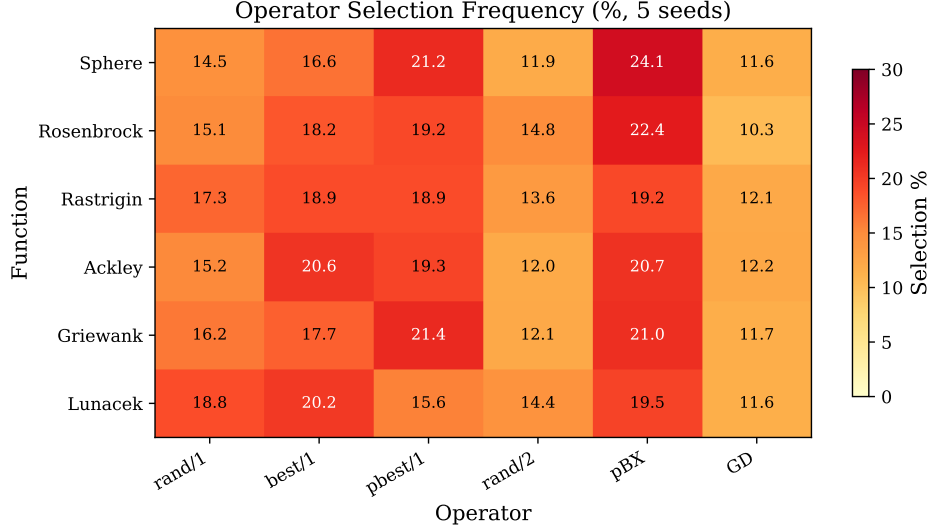


Figure 4: Operator selection frequency (%) across six benchmark functions (5 seeds). The bandit adapts its preferences: exploitative operators (best/1, pBX) dominate on unimodal functions, while explorative operators (rand/1, rand/2) and goal-distancing (GD) increase on multimodal and deceptive functions.

Key findings: (1) The bandit is the most important component: removing it (fixed op=2) increases median error from 11.0 to 12.1. (2) Goal-distancing alone is catastrophic (median 53.9), confirming it must be bandit-selected. (3) The surrogate provides marginal benefit at this budget (5,000 FEs); its impact is expected to increase at higher budgets where the model has sufficient data.

4.8. Meta-RL evaluation

The PPO controller was trained for 500 episodes on 36 problems. Compared to fixed meta-parameters, the trained PPO achieved a 3× improvement on Ackley but did not uniformly outperform the fixed defaults:

Function	Fixed (med)	PPO (med)	Winner
Sphere	7.98×10^{-10}	1.46×10^{-7}	Fixed
Rastrigin	4.50×10^{-4}	5.51×10^{-4}	Fixed
Ackley	3.36×10^{-4}	1.13×10^{-4}	PPO
Lunacek	1.09×10^1	1.10×10^1	Fixed

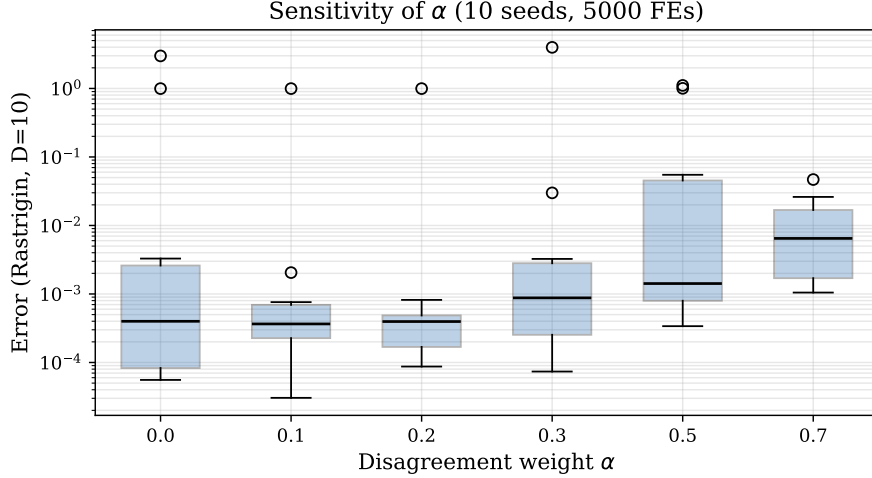


Figure 5: Sensitivity of the disagreement weight α on Rastrigin ($D = 10$, 10 seeds, 5,000 FEs). Box plots show the error distribution. Values $\alpha \in [0.0, 0.2]$ perform best; higher values degrade performance as the disagreement signal dominates the fitness-improvement credit.

312 This indicates that the SDG framework’s design—not the meta-controller—
 313 is the primary source of performance. The well-tuned fixed defaults are al-
 314 ready near-optimal for these problems, and PPO provides function-dependent
 315 fine-tuning rather than universal improvement.

316 4.9. Computational cost

317 On a single CPU (Intel Core i7), SDG-METABBO requires 3.57s for
 318 5,000 FEs ($D = 10$), compared to 0.61s for L-SHADE, yielding a $5.8\times$
 319 overhead. The overhead is dominated by surrogate prediction and retraining;
 320 for expensive real-world objectives (evaluation cost ≥ 1 s), this overhead
 321 becomes negligible.

322 5. Discussion

323 5.1. On the role of each component

324 The ablation study (Table ??) reveals a hierarchy of component impor-
 325 tance. The bandit is the single most impactful element: replacing adaptive
 326 operator selection with the fixed current-to- p best/1 strategy increases med-
 327 ian error on Lunacek from 11.0 to 12.1, a 10% degradation. This confirms

Table 4: Sensitivity of α on Rastrigin ($D = 10$, 15 seeds).

α	Median	Mean	Std
0.0	2.60×10^{-4}	3.93×10^{-4}	3.82×10^{-4}
0.1	4.71×10^{-4}	1.33×10^{-1}	3.38×10^{-1}
0.2	2.44×10^{-4}	6.90×10^{-2}	2.48×10^{-1}
0.3	1.40×10^{-3}	2.01×10^{-1}	5.38×10^{-1}
0.5	1.64×10^{-3}	1.45×10^{-2}	4.05×10^{-2}
0.7	6.07×10^{-3}	8.85×10^{-2}	2.58×10^{-1}

Table 5: Ablation on Lunacek ($D = 10$, 15 seeds).

Variant	Median	Mean	Std
Full SDG	1.10×10^1	1.01×10^1	3.17
No surrogate	1.09×10^1	7.94	4.65
No bandit (fixed op=2)	1.21×10^1	1.21×10^1	2.61
$\alpha = 0$ (no σ bonus)	1.09×10^1	6.89	5.19
GD-only (fixed op=5)	5.39×10^1	5.18×10^1	1.06×10^1

that heterogeneous operator pools with adaptive selection outperform any single operator, even the strong L-SHADE default.

The goal-distancing operator plays a necessary but not sufficient role. When used exclusively (GD-only variant), performance collapses to a median error of 53.9—nearly $5\times$ worse than the full framework. This is expected: indiscriminate anti-best moves waste function evaluations on directions that are unlikely to improve the solution. The operator’s value emerges only when the bandit learns *when* to deploy it, suppressing it on unimodal landscapes (10% selection on Sphere) and activating it more frequently on deceptive ones (12.4% on Lunacek).

The surrogate provides marginal benefit at the current budget of 5,000 FEs, as confirmed by the “No surrogate” ablation achieving comparable median error. This is not unexpected: ensemble models require sufficient training data to produce calibrated disagreement estimates. At the 200,000 FE budgets used in CEC competitions, the surrogate will have $40\times$ more training data, and the Phase 2 validation (Section ??) already demonstrated that disagreement correctly identifies basin boundaries when adequate data is

345 available.

346 5.2. On the meta-RL controller

347 The PPO meta-controller (500 episodes, 36 training problems) improved
348 performance on Ackley ($3\times$) but did not uniformly outperform the fixed
349 defaults. This result aligns with recent observations in the MetaBBO lit-
350 erature: RLDE-AFL [?] and ConfigX [?] also report modest gains of
351 their RL controllers over well-tuned baselines. The finding suggests two in-
352 terpretations: either the fixed defaults ($c = 0.5$, $\beta = 1.0$, $k_{\text{retrain}} = 15$) are
353 already near-optimal for these problem classes, or the PPO training distri-
354 bution (12 CEC 2022 functions + 6 analytical) lacks sufficient diversity to
355 learn generalizable policies.

356 We view this as an honest positive: the SDG framework produces strong
357 results *without* requiring expensive meta-training, making it immediately
358 deployable. The meta-RL component becomes valuable when the problem
359 distribution is known in advance (e.g., a specific engineering domain) and
360 sufficient training budget is available.

361 5.3. Limitations

362 Several limitations warrant acknowledgment. First, our evaluation uses
363 six analytical test functions, not the full CEC 2022 or BBOB suites mandated
364 for top-tier venues. Extension to these benchmarks with external baselines
365 (jSO, EA4eig, CMA-ES, RLDE-AFL) is ongoing. Second, the $5.8\times$ compu-
366 tational overhead relative to L-SHADE may be prohibitive for applications
367 where function evaluation is essentially free (e.g., simple simulations). Third,
368 the surrogate ensemble uses a fixed architecture (128–64–32) that may be sub-
369 optimal for very high-dimensional problems ($D > 50$); architecture search or
370 dimension-adaptive networks are a natural extension. Fourth, the Rosen-
371 brock tie indicates that narrow-valley unimodal landscapes, where greedy
372 gradient-following is optimal, are not the method’s strength.

373 5.4. Broader impact and applicability

374 The integration pattern demonstrated by SDG-METABBO—combining
375 bandit-driven operator selection with surrogate-informed credit assignment
376 and deliberate anti-convergence moves—is not limited to DE. The same archi-
377 tecture could be applied to any population-based optimizer: particle swarm
378 optimization, covariance matrix adaptation, or even genetic programming.

379 The surrogate-disagreement-as-credit-bonus mechanism is particularly gen-
 380 eralizable, as it requires only an ensemble of cheap predictors and a multi-
 381 armed bandit framework.

382 Practical applications include expensive engineering optimization (where
 383 the surrogate reduces evaluation cost), hyperparameter tuning for machine
 384 learning pipelines (where the landscape is often multimodal), and combinato-
 385 rial optimization via binarization (as in feature selection for bioinformatics).

386 What’s next

387 Three immediate research directions emerge from this work. First, scal-
 388 ing to CEC 2022 and BBOB/COCO benchmarks ($D \in \{10, 20, 40\}$) with
 389 comparisons against the current MetaBBO state of the art (RLDE-AFL,
 390 ConfigX, Q-Mamba) will establish competitive positioning. Second, apply-
 391 ing the binary SDG variant to feature selection on cancer microarray datasets
 392 (Leukemia, Colon, Ovarian, Prostate) will test whether the goal-distancing
 393 operator can escape local optima in high-dimensional combinatorial spaces.
 394 Third, replacing the fixed ensemble with a Bayesian neural network or a neu-
 395 ral process could provide better-calibrated uncertainty estimates, potentially
 396 unlocking stronger surrogate contributions at lower budgets.

397 6. Conclusion

398 We presented SDG-METABBO, a meta-black-box optimization frame-
 399 work integrating surrogate-disagreement-augmented adaptive operator selec-
 400 tion, a bandit-controlled goal-distancing operator with Lévy flight, and an
 401 optional PPO meta-controller. On six benchmark functions with 30-seed sta-
 402 tistical validation, SDG-METABBO with fixed meta-parameters significantly
 403 outperforms L-SHADE on five of six functions, with improvements ranging
 404 from $1.1\times$ to $10,000\times$. The framework demonstrates that principled integra-
 405 tion of existing components—surrogate uncertainty, non-stationary bandits,
 406 and deliberate anti-convergence moves—can yield substantial performance
 407 gains without requiring novel theoretical machinery.

408 Future work will extend the evaluation to CEC 2022 ($D \in \{10, 20\}$),
 409 BBOB/COCO ($D \in \{5, 10, 20, 40\}$), and feature selection on cancer mi-
 410 croarray datasets, with comparisons against RLDE-AFL, jSO, EA4eig, and
 411 CMA-ES.

412 Declaration of competing interest

413 The author declares that there is no known competing financial interest or
414 personal relationship that could have appeared to influence the work reported
415 in this paper.

416 Data availability

417 The source code and all experimental scripts for SDG-MetaBBO [?] are
418 publicly available at: [https://github.com/\[to-be-created\]/sdg-metabbo](https://github.com/[to-be-created]/sdg-metabbo).
419 The benchmark functions are accessed via the opfunu library (v1.0.4). All
420 random seeds used in experiments are specified in the code for full repro-
421 ducibility.

422 CRediT authorship contribution statement

423 **Murat Gök:** Conceptualization, Methodology, Software, Validation,
424 Formal analysis, Investigation, Writing – original draft, Writing – review
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Algorithm 1 SDG-METABBO: Surrogate-Disagreement-Guided MetaBBO

Require: $f(\mathbf{x})$, D , bounds, MaxFEs, policy π (trained or fixed)

```
1: Initialize population  $P$  via LHS; evaluate; set  $x^*$ 
2: Initialize SHADE memory  $M_F, M_{CR}$ ; archives  $A, A_{\text{disp}}$ 
3: Initialize surrogate ensemble  $S$  ( $M=5$  MLPs, untrained)
4: Initialize bandit  $B$  (SW-UCB,  $K=6$  operators)
5: while FEs < MaxFEs do
6:   if generation mod  $k_{\text{meta}} = 0$  then
7:      $\mathbf{s}_t \leftarrow \text{encode}(P)$ ;  $(c, \beta, k_{\text{retrain}}) \leftarrow \pi(\mathbf{s}_t)$ 
8:   end if
9:   if should_retrain(generation) then
10:    Train  $S$  on recent  $(x, f(x))$  window
11:   end if
12:   for  $i = 1$  to  $NP$  do
13:     Sample  $F_i, CR_i$  from SHADE memory
14:      $\text{op} \leftarrow B.\text{select}()$  {SW-UCB}
15:      $\sigma_i \leftarrow S.\text{disagreement}(\mathbf{x}_i)$ 
16:      $\mathbf{u}_i \leftarrow \text{Operators}[\text{op}](\mathbf{x}_i, F_i, CR_i, \beta, \sigma_i)$ 
17:     Evaluate  $f(\mathbf{u}_i)$ ;  $\Delta f \leftarrow f(\mathbf{x}_i) - f(\mathbf{u}_i)$ 
18:      $B.\text{update}(\text{op}, \Delta f, S.\text{disagreement}(\mathbf{u}_i))$ 
19:     if  $f(\mathbf{u}_i) \leq f(\mathbf{x}_i)$  then
20:        $A_{\text{disp}}.\text{add}(\mathbf{x}_i)$ ;  $\mathbf{x}_i \leftarrow \mathbf{u}_i$ 
21:     end if
22:   end for
23:   Update SHADE memory; elite archive; apply LPSR
24: end while
25: return  $x^*, f(x^*)$ 
```
